

Task 4 Debugging with AI

Prompt : Debug the following Python code.

Identify errors like syntax errors, runtime errors and logical errors.

Explain each errors.

Provide the corrected version of the code.

Give a proper explanation with easy to understand language.

Code:

```
numbers = [1,2,3,4,5]
sum = 0
for i in range(0, len(numbers)+1):
    sum += numbers[i]
print("Average:", sum/0)
```

Output:

The given code has one runtime error and one logical issue. There are no syntax errors.

Original Code

```
Python
numbers = [1,2,3,4,5]
sum = 0
for i in range(0, len(numbers)+1):
    sum += numbers[i]
print("Average:", sum/6)
```

1. Runtime Error: List Index Out of Range

Problem

```
Python
for i in range(0, len(numbers)+1):
```

- `len(numbers)` is 5 because the list has 5 elements.
- `range(0, len(numbers)+1)` generates:

```
Python
0, 1, 2, 3, 4, 5
```

- The valid indexes of the list are:

```
Python
0, 1, 2, 3, 4
```

- When `i` becomes 5, Python tries to access:

```
Python
numbers[5]
```

which does not exist, causing:

```
IndexError: list index out of range
```



Fix

Loop only until the last valid index:

Python



Run

```
for i in range(len(numbers)):
```

2. Runtime Error: Division by Zero

Problem

Python



Run

```
print("Average:", sum/0)
```

- Dividing any number by 0 is not allowed in Python.
- This causes:

```
ZeroDivisionError: division by zero
```



Fix

To calculate the average, divide by the number of elements in the list:

Python



Run

```
print("Average:", sum / len(numbers))
```



3. Logical Issue: Using `sum` as a Variable Name

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Problem

Python



Run

```
sum = 0
```

Python already has a built-in function named `sum()`.

Using `sum` as a variable hides the built-in function, which can lead to confusion or errors later in larger programs.

Better Practice

Use a different variable name, such as:

Python



Run

```
total = 0
```

Corrected Code

Python



Run

```
numbers = [1, 2, 3, 4, 5]
```

```
total = 0
```

```
for i in range(len(numbers)):
    total += numbers[i]
```

```
print("Average:", total / len(numbers))
```